ECE 329 **Homework 1 — Solutions** **Due: Thursday, September 3, 2026, 11:59:00PM**

1. Consider the 3-D vectors

$$\begin{aligned}\mathbf{A} &= \hat{x} + 3\hat{y} - 2\hat{z},\\ \mathbf{B} &= \hat{x} + \hat{y} - \hat{z},\\ \mathbf{C} &= -\hat{x} + 3\hat{y} + 2\hat{z},\end{aligned}$$

a) The vector

$$\mathbf{A} - \mathbf{B} + 3\mathbf{C} = -3\hat{x} + 11\hat{y} + 5\hat{z}$$

b) The vector *magnitude*

$$|\mathbf{A} - \mathbf{B} + 3\mathbf{C}| = \sqrt{3^2 + 11^2 + 5^2} = \sqrt{155} \approx 12.45$$

c) The unit vector

$$\hat{u} = \frac{\mathbf{A} + 2\mathbf{B} - \mathbf{C}}{|\mathbf{A} + 2\mathbf{B} - \mathbf{C}|} = \frac{4\hat{x} + 2\hat{y} - 6\hat{z}}{\sqrt{4^2 + (2)^2 + (-6)^2}} \approx 0.53\hat{x} + 0.27\hat{y} - 0.80\hat{z}$$

d) The *dot product*

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A} = (\hat{x} + 3\hat{y} - 2\hat{z}) \cdot (\hat{x} + \hat{y} - \hat{z}) = 1 \times 1 + 3 \times 1 + (-2) \times (-1) = 6$$

e) The *cross product*

$$\begin{aligned}\mathbf{B} \times \mathbf{C} = -\mathbf{C} \times \mathbf{B} &= (\hat{x} + \hat{y} - \hat{z}) \times (-\hat{x} + 3\hat{y} + 2\hat{z})\\ &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & 1 & -1 \\ -1 & 3 & 2 \end{vmatrix}\\ &= 5\hat{x} - \hat{y} + 4\hat{z}\end{aligned}$$

f) Since $\mathbf{B} \cdot \mathbf{C} = -1 + 3 - 2 = 0$, $\mathbf{B}$ and $\mathbf{C}$ are orthogonal.

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2. We know$Q_1 = -8\pi\varepsilon_0$, so $Q_2 = -Q_1/2 = 4\pi\varepsilon_0$. The electric field at the point P_3 is the superposition of those induced by Q_1 and Q_2, namely

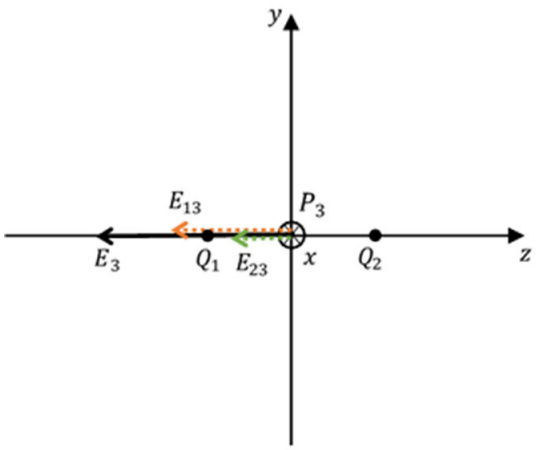
$$\begin{aligned}
\mathbf{E_3} &= \sum_{i=1}^{2} \frac{Q_i}{4\pi\varepsilon_0 |\mathbf{r}_3 - \mathbf{r_i}|^2} \cdot \frac{\mathbf{r_3} - \mathbf{r_i}}{|\mathbf{r}_3 - \mathbf{r}_i|} \\
&= (-2)\,\frac{\hat{z}}{|\hat{z}|^3} + (1)\,\frac{-\hat{z}}{|-\hat{z}|^3} \\
&= -3\hat{z}\,(V/m).
\end{aligned}$$

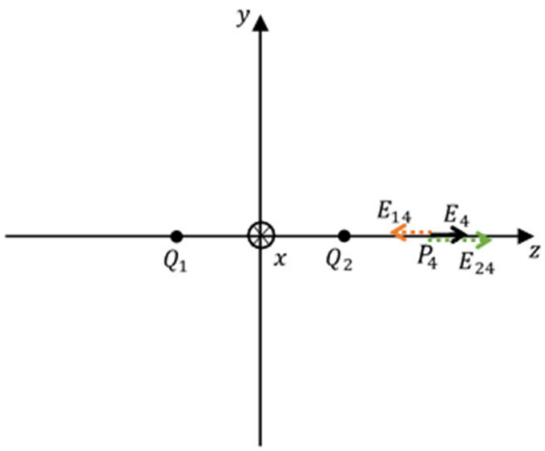
The electric field at the point P_4 and P_5 can be obtained in a similar way

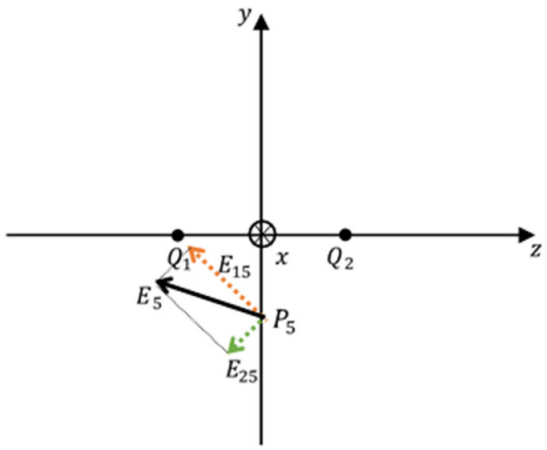
$$\begin{aligned}
\mathbf{E_4} &= \sum_{i=1}^{2} \frac{Q_i}{4\pi\varepsilon_0 |\mathbf{r}_4 - \mathbf{r_i}|^2} \cdot \frac{\mathbf{r_4} - \mathbf{r_i}}{|\mathbf{r}_4 - \mathbf{r}_i|} \\
&= (-2)\,\frac{3\hat{z}}{|3\hat{z}|^3} + (1)\,\frac{\hat{z}}{|\hat{z}|^3} \\
&\approx 0.778\hat{z}\,(V/m).
\end{aligned}$$

$$\begin{aligned}
\mathbf{E_5} &= \sum_{i=1}^{2} \frac{Q_i}{4\pi\varepsilon_0 |\mathbf{r}_5 - \mathbf{r_i}|^2} \cdot \frac{\mathbf{r_5} - \mathbf{r_i}}{|\mathbf{r}_5 - \mathbf{r}_i|} \\
&= (-2)\,\frac{-\hat{y} + \hat{z}}{|-\hat{y} + \hat{z}|^3} + (1)\,\frac{-\hat{y} - \hat{z}}{|-\hat{y} + \hat{z}|^3} \\
&= \frac{1}{2\sqrt{2}}\hat{y} - \frac{3}{2\sqrt{2}}\hat{z} \\
&\approx 0.35\hat{y} - 1.06\hat{z}\,(V/m).
\end{aligned}$$

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3. In the three cases, we have the same $\mathbf{E}$ and $\mathbf{B}$ at the origin, but different $\mathbf{v}$. Thus, we have different $\mathbf{F}$. In each case, $\mathbf{v}$ and $\mathbf{F}$ must satisfy $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. Therefore, we have the following three equations

$$\begin{cases} & 2\hat{x} - \hat{y} & = & \mathbf{E} \\ \mathbf{F_2} - \mathbf{E} & = & -2\hat{y} & = & \mathbf{v_2} \times \mathbf{B} \\ \mathbf{F_3} - \mathbf{E} & = & \hat{x} - \hat{y} - \hat{z} & = & \mathbf{v_3} \times \mathbf{B} \end{cases}$$

Assume that $\mathbf{B} = B_x\hat{x} + B_y\hat{y} + B_z\hat{z}$, we obtain

$$\begin{cases} \mathbf{E} & = & 2\hat{x} - \hat{y} \\ B_y & = & 0 \\ -2B_x\hat{y} & = & -2\hat{y} \\ (\hat{x} - \hat{y} - \hat{z}) \cdot \mathbf{B} & = & 0 \end{cases}$$

Then

$$\begin{cases} B_y & = & 0 \\ B_x & = & 1 \\ B_x - B_z & = & 0 \end{cases}$$

Thus

$$\begin{cases} B_y & = & 0 \\ B_x = B_z & = & 1 \end{cases}$$

In summary, $\mathbf{E} = 2\hat{x} - \hat{y}\,(V/m)$ and $\mathbf{B} = \hat{x} + \hat{z}\,(Wb/m^2)$.

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4.

a) Let us use S_1, S_2, S_3, S_4, S_5 and S_6 to denote the surfaces $x = 0$, $x = 2$, $y = 0$, $y = 2$, $z = 0$ and $z = 2$, respectively. We consider S_1 and S_2 first. The unit vector on S_1 pointing away from the volume is along $-\hat{x}$ direction, so we have

$$\int_{S_1} \mathbf{J} \cdot d\mathbf{S} = \int_0^2 \int_0^2 [x^2 y(\hat{x} + \hat{y} + \hat{z})] \,|_{x=0} \cdot (-\hat{x}) dy dz = 0\,(A).$$

For S_2, the unit vector pointing away from the volume is along $+\hat{x}$ direction, therefore

$$\int_{S_2} \mathbf{J} \cdot d\mathbf{S} = \int_0^2 \int_0^2 [x^2 y(\hat{x} + \hat{y} + \hat{z})] \,|_{x=2} \cdot (+\hat{x}) dy dz = 16\,(A).$$

Similarly, for S_3, S_4, S_5 and S_6, we can obtain

$$\int_{S_3} \mathbf{J} \cdot d\mathbf{S} = \int_0^2 \int_0^2 [x^2 y(\hat{x} + \hat{y} + \hat{z})] \,|_{y=0} \cdot (-\hat{y}) dx dz = 0\,(A),$$

$$\int_{S_4} \mathbf{J} \cdot d\mathbf{S} = \int_0^2 \int_0^2 [x^2 y(\hat{x} + \hat{y} + \hat{z})] \,|_{y=2} \cdot (+\hat{y}) dx dz = \frac{32}{3}\,(A),$$

$$\int_{S_5} \mathbf{J} \cdot d\mathbf{S} = \int_0^2 \int_0^2 [x^2 y(\hat{x} + \hat{y} + \hat{z})] \,|_{z=0} \cdot (-\hat{z}) dx dy = -\frac{16}{3}\,(A),$$

$$\int_{S_6} \mathbf{J} \cdot d\mathbf{S} = \int_0^2 \int_0^2 [x^2 y(\hat{x} + \hat{y} + \hat{z})] \,|_{z=2} \cdot (+\hat{z}) dx dy = \frac{16}{3}\,(A).$$

Therefore,

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \sum_{i=1}^{6} \int_{s_i} \mathbf{J} \cdot d\mathbf{S} = \frac{80}{3} \approx 26.67\,(A).$$

b) Divergence theorem followed by Continuity equation:

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{J} dV = -\frac{\partial}{\partial t} \int_V \rho dV = -\frac{\partial Q_{enclosed}}{\partial t}$$

In words, since the current density flux is positive, there is current comming out of the charged region, thus due to charge conservation, we should expect a decreasing rate (loss) of charges within it.

c) By Gauss law: *Electric flux* $\sim$ *Charges enclosed*
From (b), a decreasing of charge would lead to a decreasing of eletric flux through the enclosed surface.

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5.

a) The force experienced by the charges is only due to the Lorentz force $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. The acceleration can be found using Newton's 2nd law $\mathbf{F} = \frac{dm\mathbf{v}}{dt} = m\mathbf{a}$. Since the magnetic field is zero and the electric field is $\mathbf{E} = E_0\hat{z}$, the acceleration $\mathbf{a_e}$ and $\mathbf{a_p}$ of an electron and a proton, respectively, are given by:

$$\begin{aligned} \mathbf{a_e} &= \frac{-eE_0}{m_e}\hat{z} = \frac{-1.60219\times10^{-19}}{9.10938\times10^{-31}}E_0\hat{z} = -1.759\times10^{11}E_0\hat{z}\ \left(\frac{m}{s^2}\right) \\ \mathbf{a_p} &= \frac{eE_0}{m_p}\hat{z} = \frac{1.60219\times10^{-19}}{1.67262\times10^{-27}}E_0\hat{z} \approx 9.579\times10^{7}E_0\hat{z}\ \left(\frac{m}{s^2}\right) \end{aligned}$$

b) The position of the charges can be determined by integrating the acceleration twice with respect to time.

$$\mathbf{r} = \mathbf{r_0} + \mathbf{v_0}t + \frac{1}{2}\mathbf{a}t^2$$

Given $\mathbf{r_0} = 0$, $\mathbf{v_0} = 0$ then, $\mathbf{r} = \frac{qE_0}{2m}t^2\hat{z}$. Using $E_0 = 1\frac{mV}{m}$ and $t = 1s$ gives

$$\begin{aligned} \mathbf{r_e}(1s) &= \tfrac{1}{2}\mathbf{a_e}t^2 = \tfrac{-eE_0}{2m_e}t^2\hat{z} &&= -8.8\times10^4\hat{z}\,(km) \\ \mathbf{r_p}(1s) &= \tfrac{1}{2}\mathbf{a_p}t^2 = \tfrac{eE_0}{2m_p}t^2\hat{z} &&= 47.89\hat{z}\,(km) \end{aligned}$$

c) By definition:

$$\mathbf{J} = \lim_{A\to0}\frac{\mathbf{I}}{A} = \lim_{A\to0}\frac{dQ\hat{l}}{Adt} = \lim_{A\to0}\frac{d(\rho A\mathbf{l})}{Adt} = \rho\frac{d\mathbf{l}}{dt} = \rho\mathbf{v} = qN\mathbf{v}$$

The total current density $\mathbf{J_t}$ is obtained by adding the contribution of both charges.

$$\begin{aligned} \mathbf{J_t} &= \mathbf{J_e} + \mathbf{J_p} \\ \mathbf{J_t}(t) &= -eN_e(\mathbf{a_e}t) + eN_p(\mathbf{a_p}t) \\ &= E_0e^2N\left(\tfrac{1}{m_e} + \tfrac{1}{m_p}\right)t\hat{z} \\ &\approx 28.19t\hat{z}\ \tfrac{A}{m^2} \end{aligned}$$

Remark: This current density is not physically possible. The current density can't increase to infinity as time increases under a linear relationship as shown above. We need to include friction which indeed exists and will balance out the Lorentz force, create a steady state drift velocity and current density. Also, since a proton has a much larger mass than an electron, $m_p \gg m_e$, the term $\frac{1}{m_p}$ which represents the proton's contribution is, to a very good approximation, negligible compared to that of the electron. Plus, note that by writing $\mathbf{J}$ in terms of q and $\mathbf{v}$, we end up having a q^2 in $\mathbf{J}$. That is to say, no matter the sign of the charge carrier, $\mathbf{J}$ is always in the same direction of the excitation $\mathbf{E}$.

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6.

a) The total charge can be found from Gauss' law, $\oint_S \boldsymbol{D} \cdot d\boldsymbol{S} = Q_V$ and the displacement field definition $\boldsymbol{D} = \epsilon_0 \boldsymbol{E}$. Since the field is $\hat{x}$ oriented, the only two faces of the cube that will contribute to the integral occur at $x = \pm 1$, denoted $S_\pm$. To perform the integral, we will use the cartesian definition of $dA = dydz$.

$$\boldsymbol{D} = \epsilon_0 \boldsymbol{E} = \epsilon_0 k y^2 z^4 \cos(\pi x)\hat{x} \implies$$

$$\oint_S \boldsymbol{D} \cdot d\boldsymbol{S} = \oint_{S_+} \boldsymbol{D} \cdot \hat{x} dA + \oint_{S_-} \boldsymbol{D} \cdot (-\hat{x}) dA$$

$$= \oint_{S_+} \boldsymbol{D} \cdot \hat{x} dA - \oint_{S_-} \boldsymbol{D} \cdot \hat{x} dA$$

$$= \oint_{S_+} \epsilon_0 k y^2 z^4 \cos(\pi) dydz - \oint_{S_-} \epsilon_0 k y^2 z^4 \cos(-\pi) dydz$$

$$= \epsilon_0 k (- \oint_{S_+} y^2 z^4 dydz + \oint_{S_-} y^2 z^4 dydz)$$

The integrals over $S_\pm$ here are the same, as we are integrating an argument independent of x over identical surfaces independent of x. This means that the integral, thus the total charge, is zero, or $Q_V = 0$.

b) A similar approach as before can be taken, where only $S_\pm$ must be handled:

$$\boldsymbol{D} = \epsilon_0 \boldsymbol{E} = \epsilon_0 k y^2 z^4 e^x \hat{x} \implies$$

$$\oint_S \boldsymbol{D} \cdot d\boldsymbol{S} = \oint_{S_+} \boldsymbol{D} \cdot \hat{x} dA + \oint_{S_-} \boldsymbol{D} \cdot (-\hat{x}) dA$$

$$= \oint_{S_+} \boldsymbol{D} \cdot \hat{x} dA - \oint_{S_-} \boldsymbol{D} \cdot \hat{x} dA$$

$$= \oint_{S_+} \epsilon_0 k y^2 z^4 e^1 dydz - \oint_{S_-} \epsilon_0 k y^2 z^4 e^{-1} dydz$$

$$= \epsilon_0 k (e^1 - e^{-1}) \int_{-1}^{1} \int_{-1}^{1} y^2 z^4 dydz$$

$$= 2\epsilon_0 k \sinh(1) \int_{-1}^{1} y^2 dy \int_{-1}^{1} z^4 dz$$

$$= 2\epsilon_0 k \sinh(1) (\frac{2}{3})(\frac{2}{5})$$

$$= \frac{8}{15} \epsilon_0 k \sinh(1) = Q_V$$

$$Q_V \approx 1.664 \times 10^{-11} \text{C}$$

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